

EMPLOYEE DATA SYNCHRONIZATION

Technical Project Documentation

Integration: SuccessFactors → S/4HANA & S/4HANA → SFTP Payroll

Document Attribute	Value
Prepared By	Vinod Mudavath
Role	SAP Integration Developer
Trainer	Laxmi Ahuja
Business Context	SAP Integration Suite Capstone Project
Document Version	1.0
Status	Final Project / Technical Handover
Date	14 August 2026

DOCUMENT PURPOSE

This document explains the completed integration from business requirement through architecture, configuration, transformation, exception handling, security, testing, monitoring, operational support and evidence. It is structured so that a reviewer, developer, tester or support engineer can understand the solution without relying only on the iFlow canvas.

Document Control & Approval

Item	Value	Owner / Note
Document title	Employee Data Synchronization — Technical Project Documentation	Vinod Mudavath
Project	Integration: SuccessFactors → S/4HANA & S/4HANA → SFTP Payroll	Vinod Mudavath
Version	1.0	Initial client-ready handover
Status	Final Project Documentation	Prepared after implementation
Environment	SAP Integration Suite / Cloud Integration training-capstone landscape	As evidenced by supplied screenshots
Prepared by	Vinod Mudavath	SAP Integration Developer
Trainer	Laxmi Ahuja	Training / review reference
Approval	To be completed by reviewer/client	Signature/date placeholder

Executive Summary

The Employee Data Synchronization project demonstrates an end-to-end integration scenario using SAP Integration Suite / SAP Cloud Integration. Employee master data is retrieved from SAP SuccessFactors through OData V2, transformed into the required S/4HANA structure, posted to SAP S/4HANA through OData, and then prepared for downstream payroll file exchange through XML-to-CSV conversion and SFTP delivery.

The solution also contains an Exception Subprocess. When message processing fails, the error path captures the exception, prepares an administrator notification and sends the notification through the configured mail channel. This separates business processing from operational alerting and improves supportability.

The supplied implementation evidence contains CPI/iFlow configuration screens, mapping screens, runtime/configuration evidence, CSV/SFTP evidence, mail evidence and presentation/demo evidence. The screenshots are retained in Appendix A as an audit-style implementation record.

CLIENT HANDOVER VIEW

The important outcome is not only the iFlow canvas. The handover consists of the business requirement, technical design, field contract, connectivity model, error strategy, test evidence, monitoring procedure and support guidance needed to operate or enhance the integration.

Business Requirement

GlobalTech Ltd. requires reliable movement of employee information from its HR source to the ERP/master-data system and onward to a third-party payroll process. The integration should reduce manual movement, maintain consistent employee values and provide an operational notification when processing fails.

Problem Statement

- Manual movement of employee master data can introduce delays and duplicate effort.
- Different systems can contain inconsistent representations of employee attributes.
- File-based payroll handoffs require a predictable CSV structure and delivery location.
- Operational failures need to be visible to administrators instead of silently stopping the process.
- Support teams require traceability through message monitoring and runtime evidence.

Project Objectives

- Retrieve employee master data from SuccessFactors using OData V2.
- Transform source attributes into the target S/4HANA employee structure.
- Send employee data to the configured S/4HANA OData service.
- Retrieve/process employee records for downstream payroll preparation.
- Convert the required XML representation into CSV.
- Deliver the payroll-oriented CSV through SFTP.
- Implement exception handling and administrator mail notification.
- Provide monitoring and traceability through SAP Cloud Integration.

In Scope

- SuccessFactors OData V2 connectivity and employee retrieval.
- Message transformation and mapping.
- General Splitter / record-level processing.
- S/4HANA OData request-reply processing.
- Filtering and root-node preparation for downstream file generation.
- XML-to-CSV conversion and SFTP delivery.
- Exception Subprocess and mail notification.
- Testing, monitoring and implementation evidence.

Out of Scope / Not Evidenced

- Production deployment governance, transport management and enterprise change approval are not demonstrated by the supplied evidence.
- Actual payroll-system consumption beyond SFTP delivery is not demonstrated.
- Production credentials, secrets and passwords are intentionally excluded from this document.
- Formal business sign-off and SLA/OLA commitments require client approval.

Solution Architecture

The solution uses SAP Cloud Integration as the central orchestration layer between SuccessFactors, S/4HANA, SFTP and mail notification. The architecture separates source retrieval, transformation, target synchronization, payroll file preparation and exception notification.

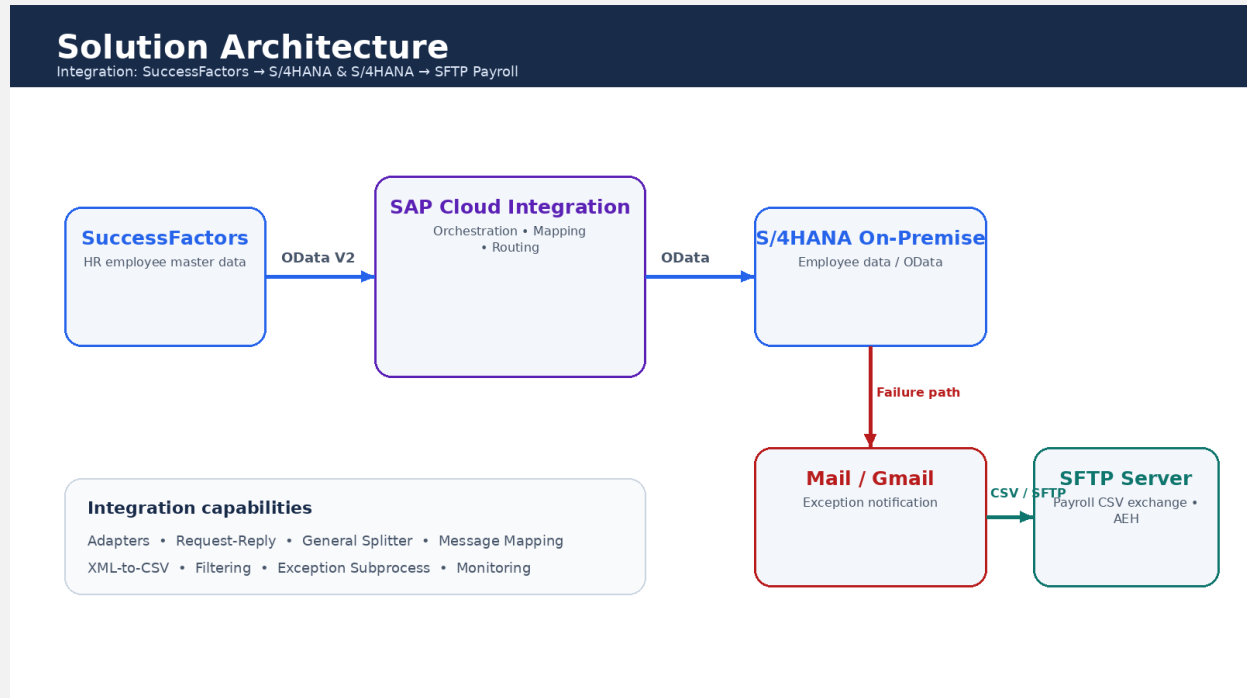


Figure 1. Logical solution architecture derived from the implemented project scope.

System Responsibilities

System / Component	Responsibility	Interface / Role
SAP SuccessFactors	Source of employee master data	OData V2
SAP Cloud Integration	Orchestration, mapping, routing, transformation, exception handling and monitoring	Central integration layer
SAP S/4HANA On-Premise	Employee data target/system of record	OData / service ZEMPDET_SRV_SRV
SFTP Server	Payroll-oriented CSV file exchange	SFTP directory AEH
Mail / Gmail	Operational exception notification	Mail adapter / SMTP setup

Architecture Principles

- Centralize integration logic in SAP Cloud Integration.
- Keep credentials in security material / credential artifacts rather than embedding passwords in the iFlow.
- Keep transformation rules explicit and traceable through message mapping.
- Separate the business path from the exception path.
- Make downstream file output deterministic through a defined field order and filename convention.
- Use monitoring and trace evidence to support troubleshooting.

05 | IFLOW DESIGN

Consolidated iFlow — End-to-End Design

The completed project uses a consolidated integration process covering employee synchronization, payroll file preparation/delivery and exception notification. The business path starts with a timer, retrieves SuccessFactors data, processes and maps employee records, posts to S/4HANA, retrieves/filters downstream data, prepares the XML structure, converts it to CSV, delivers the file to SFTP and reaches the end event.

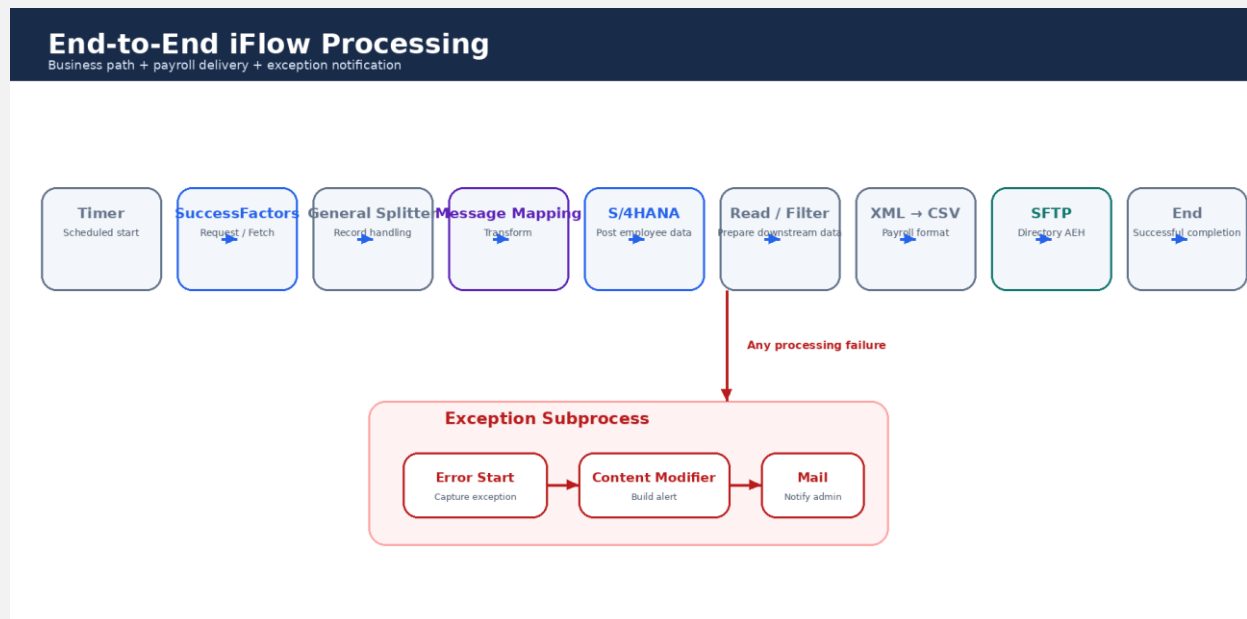
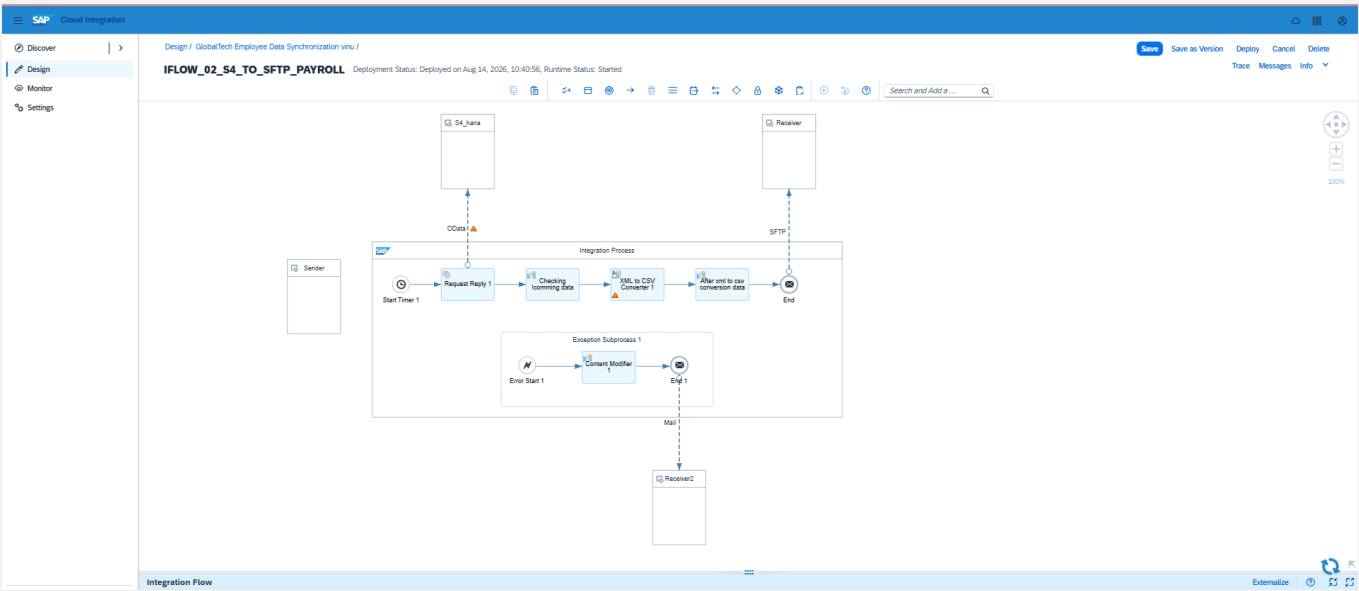
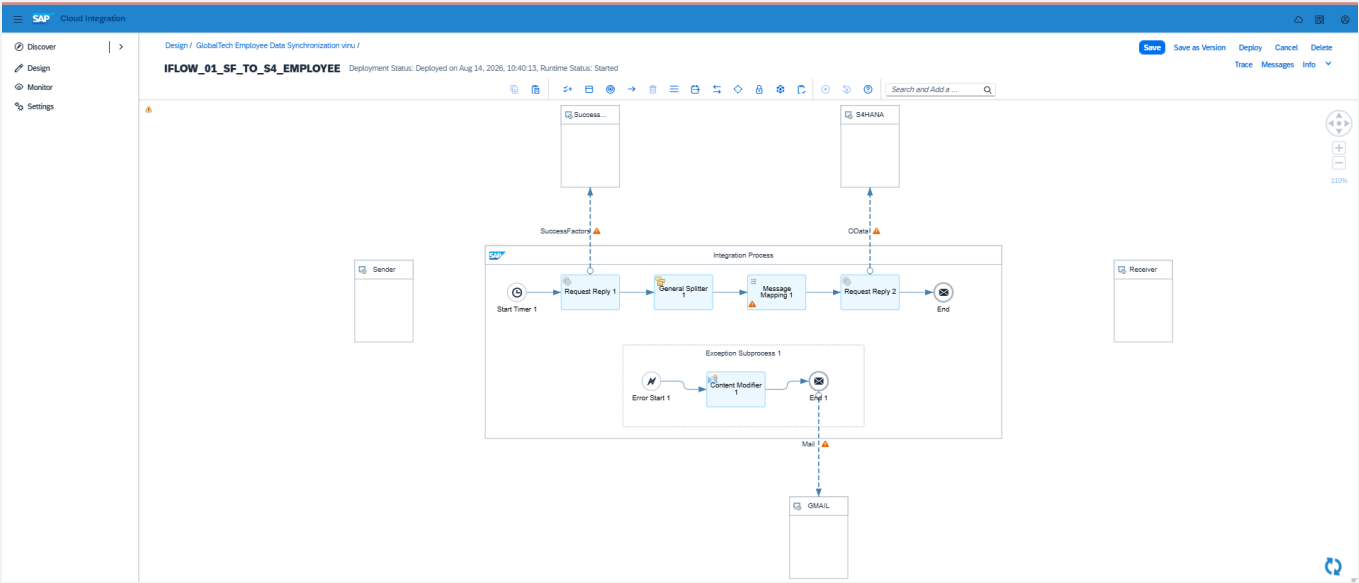


Figure 2. End-to-end logical iFlow view for the completed solution.

iFlows Diagram's



Processing Sequence

Seq.	Step	Purpose	Outcome
01	Timer	Starts scheduled processing	Repeatable execution
02	SuccessFactors Request-Reply	Reads employee master data	Source payload available
03	General Splitter	Handles employee records	Record-level processing
04	Message Mapping	Creates target employee structure	S/4HANA-compatible payload
05	S/4HANA Request-Reply	Posts transformed employee data	Target synchronization
06	Read / Filter	Retrieves and prepares downstream data	Payroll candidate records
07	Root Node Preparation	Provides predictable XML structure	Converter-ready XML
08	XML-to-CSV	Creates payroll-oriented CSV	File payload
09	SFTP	Delivers CSV to AEH	Downstream file handoff
10	End	Completes successful path	Successful completion

CONTROL POINT

A failure at a processing step is routed to the Exception Subprocess rather than being treated as a successful business completion.

06 | DETAILED IMPLEMENTATION

Step-by-Step iFlow Implementation

6.1 Timer Trigger

The timer initiates scheduled processing. The exact schedule value is environment-specific and should be confirmed from the runtime configuration before production use.

6.2 SuccessFactors OData V2

A request-reply interaction retrieves employee information. The documented source attributes include userId, firstName, lastName, country and city. Training user IDs evidenced in the project are rharrison and szhou.

6.3 General Splitter

The splitter separates employee records when the SuccessFactors response contains multiple employee entries. This supports individual-record transformation and target processing.

6.4 Message Transformation & Mapping

Message Mapping converts the SuccessFactors structure into the target Employee structure. Static client data and concatenation logic are applied as defined in the mapping contract.

6.5 S/4HANA OData Request-Reply

The transformed employee record is posted to the configured S/4HANA OData service. The documented service is ZEMPDET_SRV_SRV and the target entity is EmpdetailsSet.

6.6 Record Retrieval & Filtering

The flow retrieves employee data and applies filtering before downstream file preparation. The exact filter expression should be treated as configuration evidence and validated against the deployed iFlow.

6.7 Root Node Preparation

A root node is added to provide a predictable XML structure for conversion.

6.8 XML-to-CSV Conversion

The XML representation is converted to a CSV representation suitable for external payroll/file processing.

6.9 SFTP Processing

The generated CSV is delivered to the configured SFTP server. The documented training directory is AEH.

6.10 Success Completion

After the downstream delivery path completes successfully, the main flow reaches the end event.

07 | DATA CONTRACT & MESSAGE MAPPING

Message Mapping — SuccessFactors to S/4HANA

The target employee structure is designed around five business fields. The mapping is deterministic: Mandt is a constant, employee identity fields are constructed from first and last name, and country/city are passed through from the source.

Target Field	Source Field	Mapping Logic	Business Meaning
Mandt	—	Constant "500"	Static SAP client
Participantid	firstName + lastName	Concatenate with space	Employee full name
Country	country	Direct mapping	Uppercase if required
Sfempname	firstName + lastName	Concatenate with space	Employee full name
City	city	Direct mapping	Copied from SuccessFactors

Target Data Contract

Field	Expected Role	Mandatory / Notes
Mandt	SAP client	Constant 500
Participantid	Employee identifier / full name	firstName + space + lastName
Country	Employee country	Direct source mapping; uppercase if required
Sfempname	Employee display name	firstName + space + lastName
City	Employee city	Direct source mapping

Sample Business Output

Mandt	Participantid	Country	Sfempname	City
500	rharrison	US	Richard Harrison	Miami
500	Vinod	US	Mudavath	Dallas

MAPPING VALIDATION RULE

Participantid and Sfempname must remain consistent for the same employee. A missing firstName/lastName can prevent a required target value from being produced, so source XSD validity and target occurrence constraints must be checked before runtime testing.

08 | PAYROLL FILE & SFTP

S/4HANA → SFTP Payroll Distribution

The second business outcome is a payroll-oriented file handoff. Employee information is retrieved from S/4HANA, validated/filtered as required, prepared into a predictable XML structure, converted to CSV and transferred to the SFTP training directory.

Configuration Item	Documented Value	Purpose
SFTP directory	AEH	Payroll-oriented downstream exchange
File naming convention	<YourFirstName><YourLastName>.csv	Assessment / training naming convention
File format	CSV	Downstream file representation
Field order	Mandt, Participantid, Country, Sfempname, City	Expected business structure

File Delivery Controls

- Validate the XML structure before conversion.
- Confirm CSV headers and field order.
- Confirm file naming convention before transfer.
- Confirm SFTP directory permissions and connectivity.
- Verify the file exists in the target directory after successful execution.
- Use message monitoring to correlate the file delivery with the originating run.

09 | EXCEPTION HANDLING

Exception Handling & Notification

The exception path is intentionally isolated from the main business path. A processing exception enters the Exception Subprocess, passes through a Content Modifier and reaches the configured mail notification step.

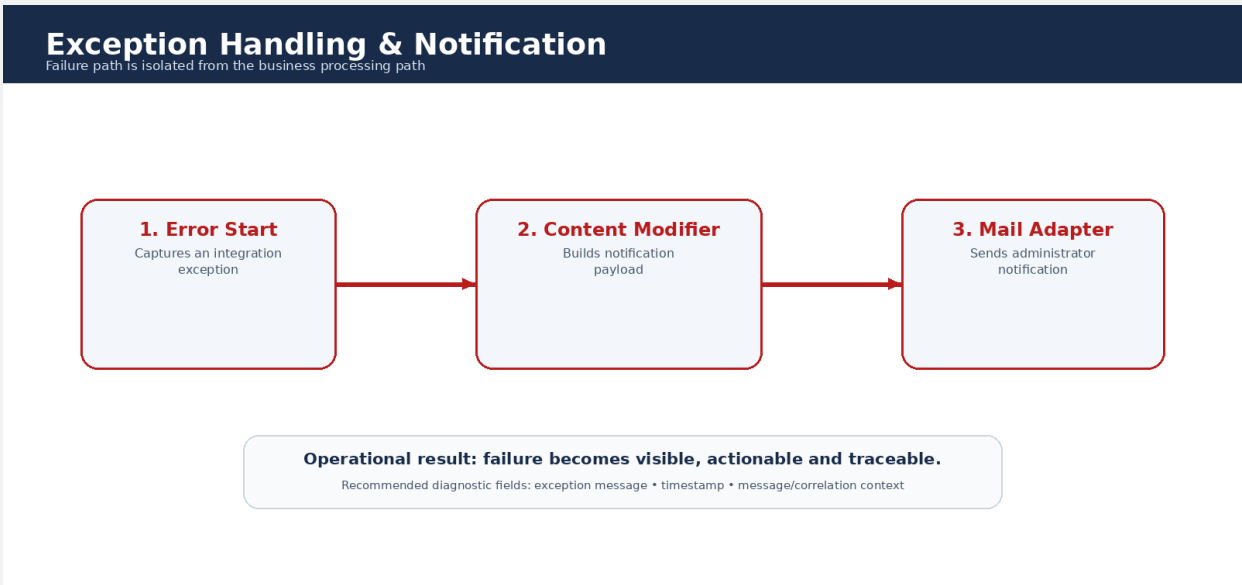


Figure 3. Logical exception-handling sequence.

Component	Responsibility	Operational Value
Error Start	Captures an integration exception	Provides controlled entry into error processing
Content Modifier	Builds notification content using available error/context information	Creates actionable alert content
Mail Adapter	Sends administrator notification	Makes failures visible to support/admin users

Recommended Notification Content

- Integration flow name / identifier.
- Timestamp of failure.
- Exception message.
- Message or correlation context where available.
- Processing step or endpoint where the error occurred.
- Recommended next action for the support user.

SECURITY NOTE

Do not place passwords, API secrets or private keys in the email body, screenshots, source control or this documentation. Use credential/security artifacts and refer to them by controlled artifact name only.

10 | SECURITY & CONNECTIVITY

Security Material & Credential Artifacts

The project evidence references credential artifacts rather than embedding passwords in the iFlow. The artifact names below are included for configuration traceability only.

Connection	Artifact / Account	Protocol / Adapter	Purpose
SuccessFactors	SFCred_Laxmi	OData V2	Read employee master data
S/4HANA	L4C_Cred	OData	Post/read employee data
SFTP	Personal training account (SFTP_vinu)	SFTP	Payroll CSV exchange
Mail	Personal training account (vinumail)	Mail / SMTP	Exception notification

Security Controls

- Keep passwords and secrets in SAP Cloud Integration Security Material.
- Use least-privilege credentials for each endpoint.
- Do not expose credentials in screenshots or documents.
- Rotate training credentials when required by the environment.
- Use separate credentials for development/test/production landscapes where applicable.
- Review certificate/host-key requirements for SFTP before production use.
- Validate endpoint trust and connectivity before deployment.

Technical Configuration Inventory

Area	Documented Configuration	Validation Required Before Production
SuccessFactors	OData V2; User entity; fields userId, firstName, lastName, country, city	Confirm tenant URL, query and field availability
S/4HANA	Service ZEMPDET_SRV_SRV; entity EmpdetailsSet; credential L4C_Cred	Validate metadata, entity operation and field types
Mapping	Mandt=500; name concatenation; country/city direct	Validate XSD occurrence and source payload
SFTP	Directory AEH; CSV file	Validate host, port, authentication, host key and permissions
Mail	Configured training/personal mail channel	Validate SMTP/receiver settings and sender policy
Monitoring	Message Monitor / trace	Validate support access and retention

Environment Parameters That Must Be Confirmed

- SuccessFactors tenant base URL.
- S/4HANA host/port and exact OData endpoint.
- SFTP host, port, directory and host-key configuration.
- Mail server/SMTP host, port and sender/receiver addresses.
- Timer schedule and timezone.
- Runtime retention/monitoring policy.

DOCUMENTATION BOUNDARY

The supplied project artifacts intentionally do not expose passwords or other secrets. Endpoint details that were not explicitly evidenced are listed as validation items rather than guessed values.

Source and Target Data Model

The source employee payload is represented around the SuccessFactors User structure. The target Employee structure contains the business fields required by the project. Message Mapping is responsible for converting between the structures.

Source Element	Target Element	Transformation	Notes
firstName + lastName	Participantid	Concatenate with space	Required employee identity value
country	Country	Direct	Uppercase if required by target
firstName + lastName	Sfempname	Concatenate with space	Display name
city	City	Direct	Source city
—	Mandt	Constant 500	Static SAP client

Validation Rules

- Source XML must conform to the source XSD used by the message mapping.
- Target required elements must receive at least one value.
- If a target element is mandatory, mapping queues/contexts must produce a value for every processed record.
- When multiple employee records are processed, context handling must preserve record boundaries.
- The resulting XML must be structurally valid before XML-to-CSV conversion.

Test Strategy

Testing should demonstrate connectivity, functional correctness, transformation accuracy, downstream delivery and negative-path behavior. The supplied evidence supports these test categories through configuration, monitoring, SFTP and mail screenshots.

Test ID	Scenario	Expected Result	Evidence to Retain
TC01	SuccessFactors connectivity	Source request succeeds	CPI message log / trace
TC02	rharrison retrieval	Expected employee record retrieved	Payload / trace
TC03	szhou retrieval	Expected employee record retrieved	Payload / trace
TC04	Message Mapping	All required target fields populated	Mapped payload / mapping test
TC05	S/4HANA processing	Target entity accepts transformed employee record	Response + message status
TC06	CSV conversion	Valid CSV generated with required header/order	Message body / file
TC07	SFTP upload	CSV written to AEH	SFTP listing
TC08	Exception path	Administrator receives error notification	Mail inbox + CPI error log

Acceptance Criteria

- Employee records can be retrieved from SuccessFactors.
- Required target fields are populated according to the mapping contract.
- S/4HANA processing completes successfully for valid records.
- CSV output follows the documented field order and naming convention.
- CSV is delivered to the configured SFTP directory.
- A controlled failure results in an administrator notification.
- The execution can be traced through SAP Cloud Integration monitoring.

EVIDENCE STATUS

The screenshots supplied with the project are retained in the evidence appendix. Where an exact runtime value or final business sign-off is not visible in the source material, this document does not invent it; the item is marked for confirmation.

Monitoring & Traceability

Operational monitoring is performed through SAP Cloud Integration message monitoring and, when required, trace/debug evidence. The SFTP directory and notification mailbox provide downstream confirmation points.

Monitoring Point	What to Check	Action if Failed
Message Monitor	Processing status, timestamps, message identifiers	Open message details and identify failed step
Trace / Payload	Source/target payload and transformation behavior	Validate mapping, context and data quality
S/4HANA response	HTTP/OData response and application error	Check endpoint, credentials and target data
SFTP directory	Expected CSV exists and is readable	Check connectivity, permissions and filename
Notification mailbox	Error alert received	Review exception content and correlate with message

Operational Runbook

- 1. Identify the failed message in Message Monitor.
- 2. Record timestamp, message ID/correlation context and failed processing step.
- 3. Determine whether the issue is connectivity, authentication, data quality, mapping, target application or file delivery.
- 4. For mapping/data issues, inspect the source payload and target structure.
- 5. For endpoint issues, validate the relevant credential artifact and endpoint configuration.
- 6. For SFTP issues, validate host key, authentication, directory and permissions.
- 7. Correct the configuration/data issue according to change-control policy.
- 8. Reprocess or rerun the message only after the cause is understood.
- 9. Verify S/4HANA acceptance, CSV creation, SFTP delivery and notification behavior as applicable.
- 10. Record the resolution in the defect/change log.

Agile / SDLC Summary

The project follows an iterative delivery lifecycle. The supplied presentation identifies the broad lifecycle as Requirement Analysis → Design → Development → Testing → Deployment → Maintenance, with sprint themes around foundation, iFlow implementation and finalization.

The SDLC framework provides a systematic process for planning, designing, developing, testing, deploying, and maintaining the integration solution.



Stage / Sprint	Activities	Output
Foundation	Requirement analysis, landscape, credentials, project setup	Confirmed scope and connectivity plan
iFlow 1	SuccessFactors retrieval, splitter, mapping, S/4HANA posting	Employee synchronization path
iFlow 2 / Payroll	S/4HANA read, filtering, XML-to-CSV, SFTP	Payroll CSV delivery path
Finalization	Exception handling, testing, monitoring, evidence, documentation	Completed handover package

Agile Artifacts

- Product / requirement backlog.
- Sprint backlog and task tracking.
- Development evidence and configuration screenshots.
- Test cases and test results.
- Defect log and resolution notes.

- Sprint review / demonstration.
- Retrospective and future improvement items.

CAPSTONE / CLIENT DISTINCTION

The Agile structure is documented here as an industrial delivery model. If the formal training submission requires specific sprint dates, story IDs, story points, burndown charts or retrospective comments, those values should be populated from the actual project records rather than estimated.



Key Technical Challenges Observed During Implementation

Challenge	Impact	Resolution / Engineering Response
Source/target XSD and occurrence alignment	Mapping may fail when required target elements do not receive values	Validate source XML against source XSD and ensure required target fields are mapped
Queue/context handling with multiple employees	Message Mapping can report insufficient values	Use record-level processing/context-aware mapping and validate splitter output
SFTP connectivity / host-key validation	File delivery can fail before transfer	Validate hostname, port, authentication and accepted host key in the environment
CSV structure preparation	Malformed XML can prevent conversion	Add predictable root structure and validate XML before conversion
Exception visibility	Failures can be hard to diagnose without context	Use Exception Subprocess + Content Modifier + Mail notification

Defect Resolution Method

- Reproduce the error with a controlled test payload.
- Identify the exact failing iFlow step.
- Inspect message body/headers and mapping context where appropriate.
- Correct configuration or transformation logic.
- Retest the original failure scenario.
- Execute a regression test on the end-to-end flow.
- Capture evidence and update the documentation.

Solution Outcome

Capability	Outcome
Employee synchronization	SuccessFactors employee data is transformed and sent to S/4HANA through the integration layer.
Transformation	Business mapping is explicit and repeatable.
Payroll file preparation	Employee data is converted into a CSV representation for downstream exchange.
SFTP delivery	Payroll-oriented file is delivered to the documented SFTP directory AEH.
Exception handling	Failures are routed to a controlled notification path.
Monitoring	Message Monitor and trace provide operational visibility.

Business Benefits

- Reduced manual employee data movement.
- Consistent transformation between HR and ERP structures.
- Predictable payroll file output.
- Improved operational visibility when integration failures occur.
- Centralized integration logic and monitoring.
- A maintainable foundation for future API/file integrations.

Recommended Future Enhancements

- Move environment-specific values into externalized parameters.
- Introduce separate DEV/TEST/PROD credential artifacts and endpoints.
- Add automated data-quality validation before S/4HANA posting.
- Add stronger duplicate/idempotency controls if required by the business.
- Introduce standardized structured error payloads and correlation IDs.
- Add alert severity, support routing and escalation rules.
- Implement automated regression test cases for mapping and file generation.
- Add transport/change-management documentation for enterprise deployment.
- Integrate with an enterprise payroll endpoint beyond the training SFTP exchange.
- Define operational SLAs, ownership, support hours and escalation procedures.

19 | CLIENT HANDOVER CHECKLIST

Final Project / Handover Package

Deliverable	Status / Evidence	Owner
Technical Project Documentation	This document	Vinod Mudavath
Agile Summary	Included within documentation / separate copy as required	Vinod Mudavath
Final PPT presentation	Prepared project presentation	Vinod Mudavath
Exported iFlow / package ZIP	To be attached to final submission	Vinod Mudavath
Test evidence	Screenshots in Appendix A	Vinod Mudavath
CSV output evidence	SFTP / file evidence screenshots	Vinod Mudavath
Exception email evidence	Mail screenshot in Appendix A	Vinod Mudavath
Client / trainer approval	Pending signature / confirmation	Reviewer

Glossary

Term	Meaning
SAP Cloud Integration / CPI	SAP Integration Suite capability used to design and run integration flows.
iFlow	Integration flow containing the orchestration and processing steps.
OData	Protocol used for service-based access to business data.
Request-Reply	Integration pattern used to send a request and wait for a response.
General Splitter	Step used to split a message into individual records/parts.
Message Mapping	Graphical transformation used to map source fields to target fields.
XML-to-CSV Converter	Transformation that converts XML data into CSV output.
SFTP	Secure file transfer protocol used for payroll file exchange.
Exception Subprocess	Dedicated error-handling path within an integration flow.
Security Material	Managed credentials/certificates used by integrations.
Message Monitor	Runtime area used to inspect integration processing status.

Source / Evidence Basis

This document is based on the supplied capstone project presentation, the earlier Agile Summary document and the 60 implementation screenshots provided with the project. Project-specific names, mapping rules, service names, directories and test scenarios are preserved from those artifacts. Items not evidenced by the source material are explicitly described as validation or approval items.

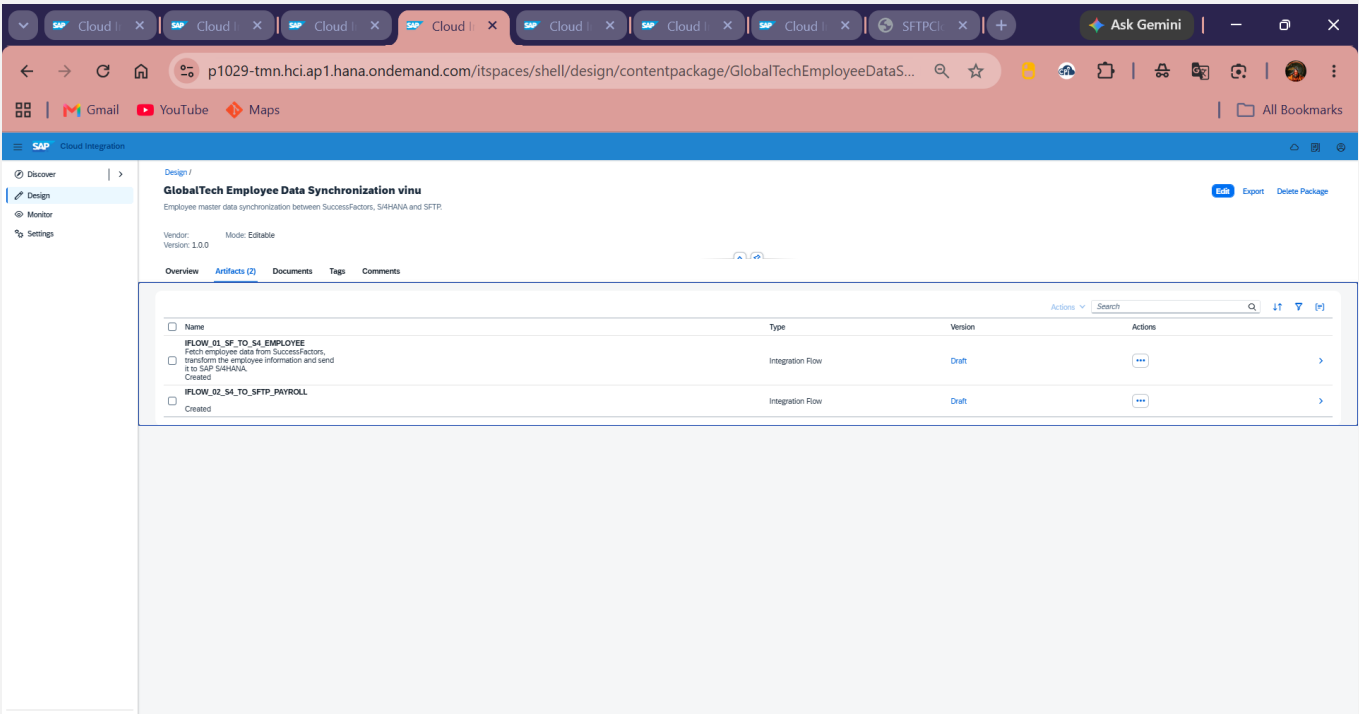
IMPORTANT

For a real client production handover, replace training-environment references with approved production landscape values, attach the approved interface specification, add formal security/architecture approvals and record actual deployment/version IDs.

Implementation Evidence — Screenshots

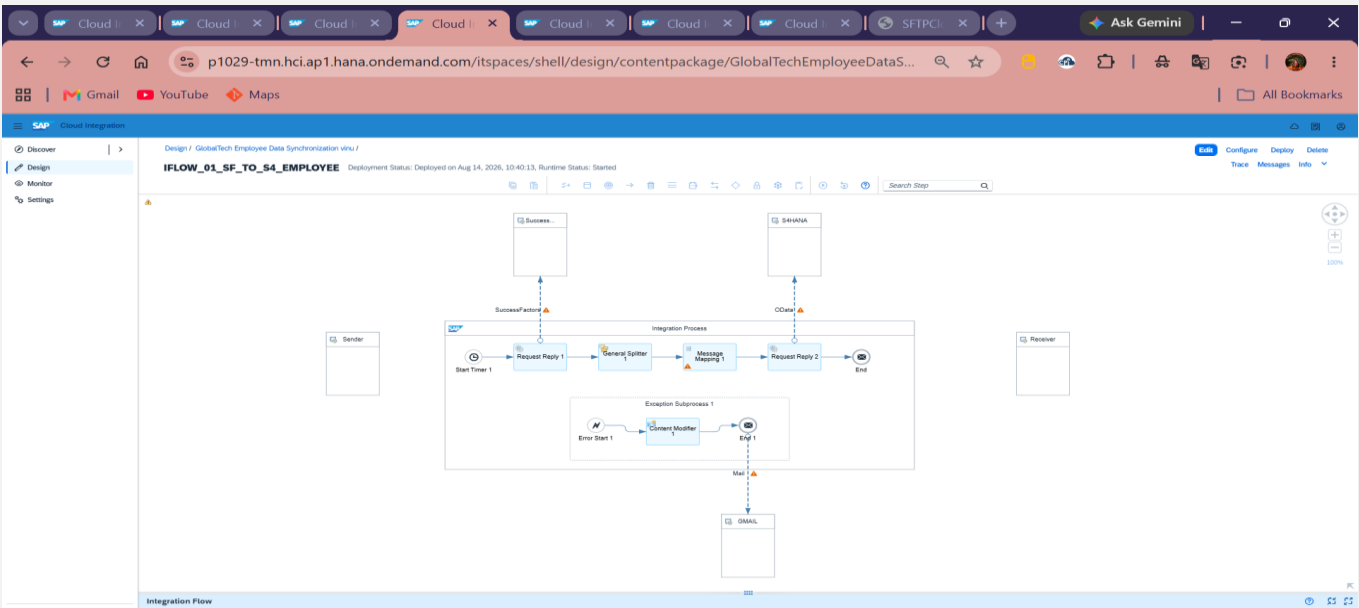
The following evidence pages preserve the supplied screenshots as a traceable record of the implementation. Each image is labeled with its evidence number so a reviewer can reference it from test or review discussions.

Evidence 01



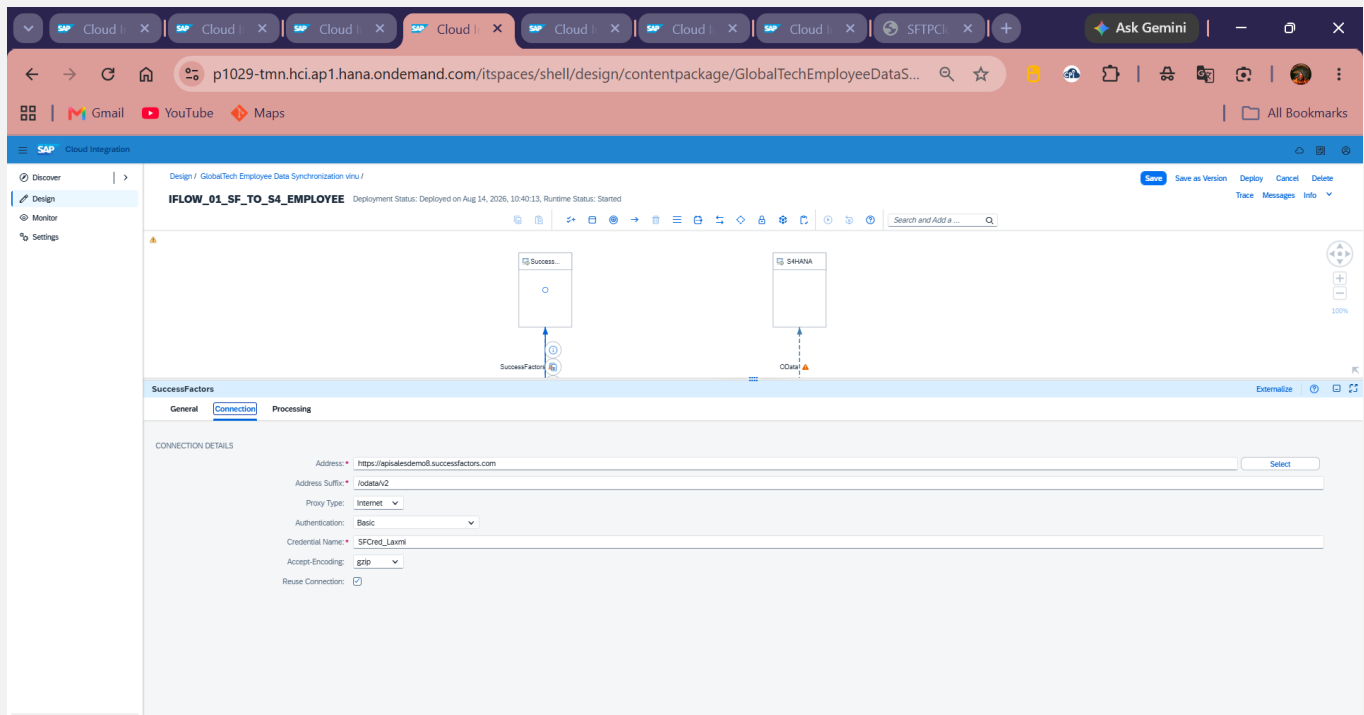
Implementation evidence screenshot 01.

Evidence 02



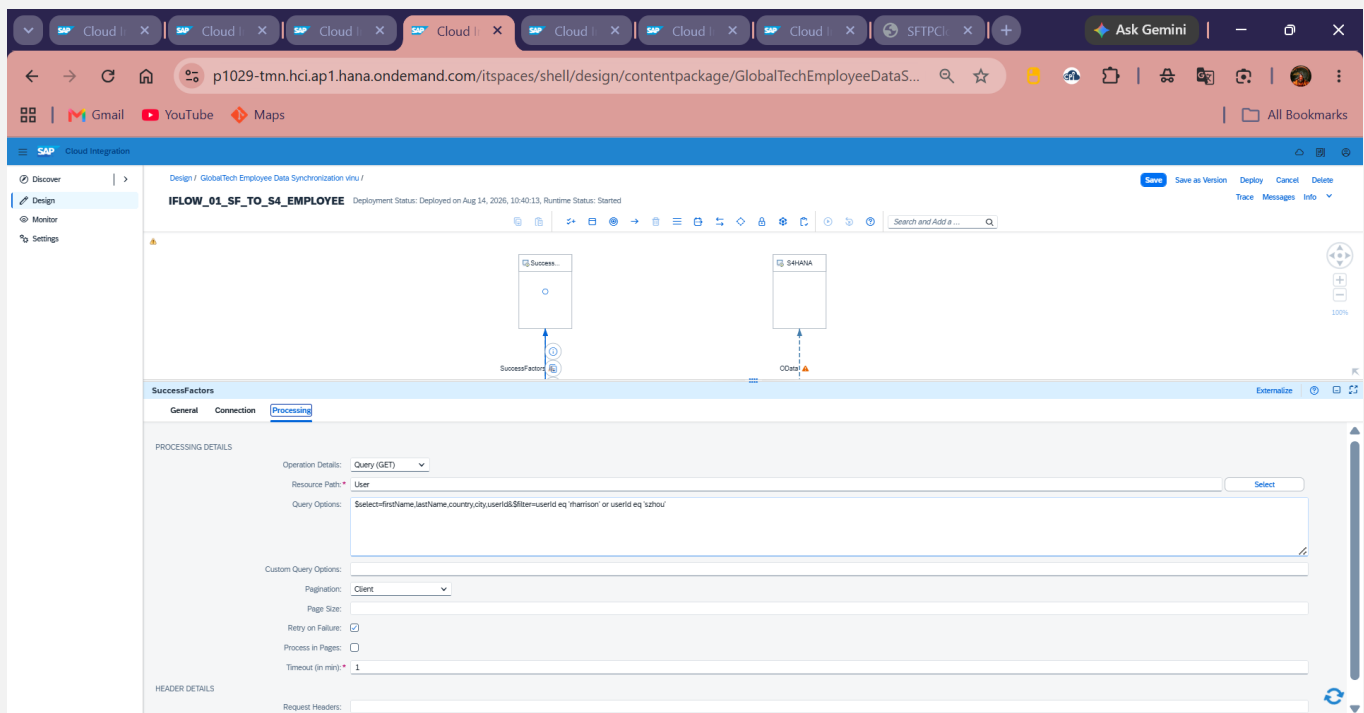
Implementation evidence screenshot 02.

Evidence 03



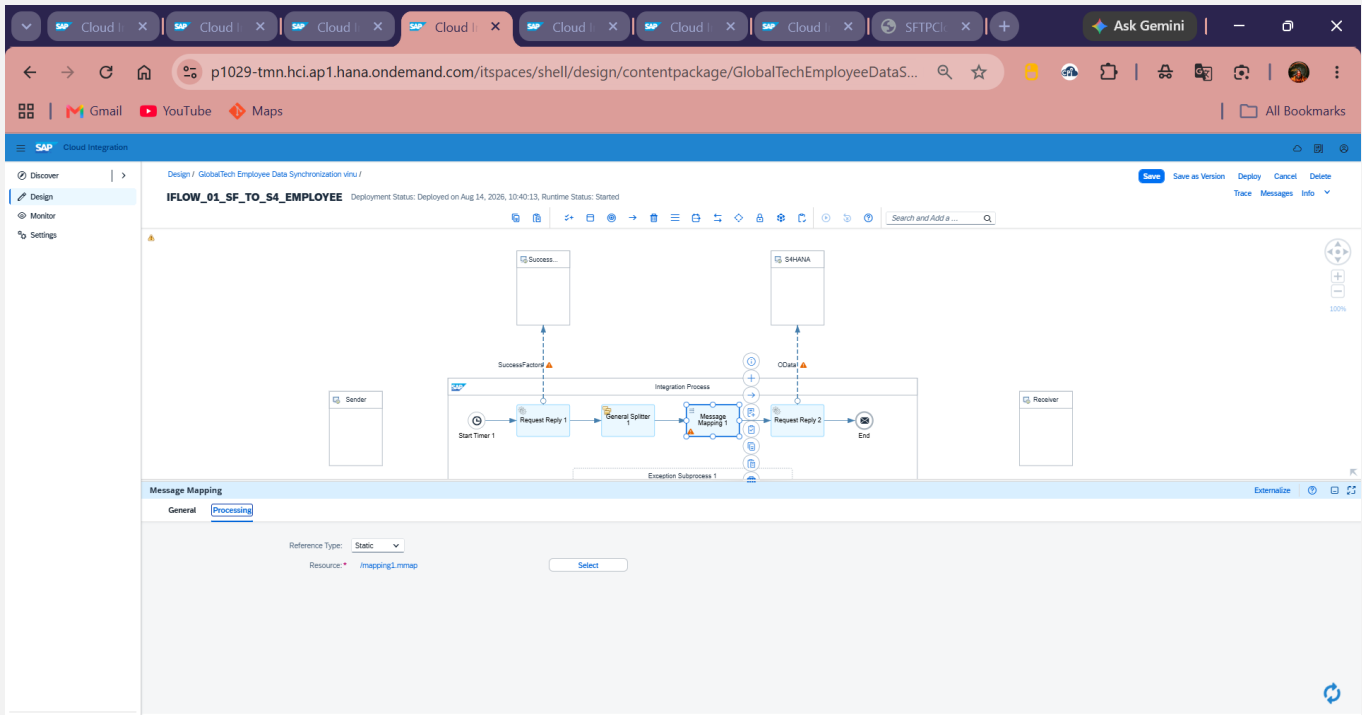
Implementation evidence screenshot 03.

Evidence 04



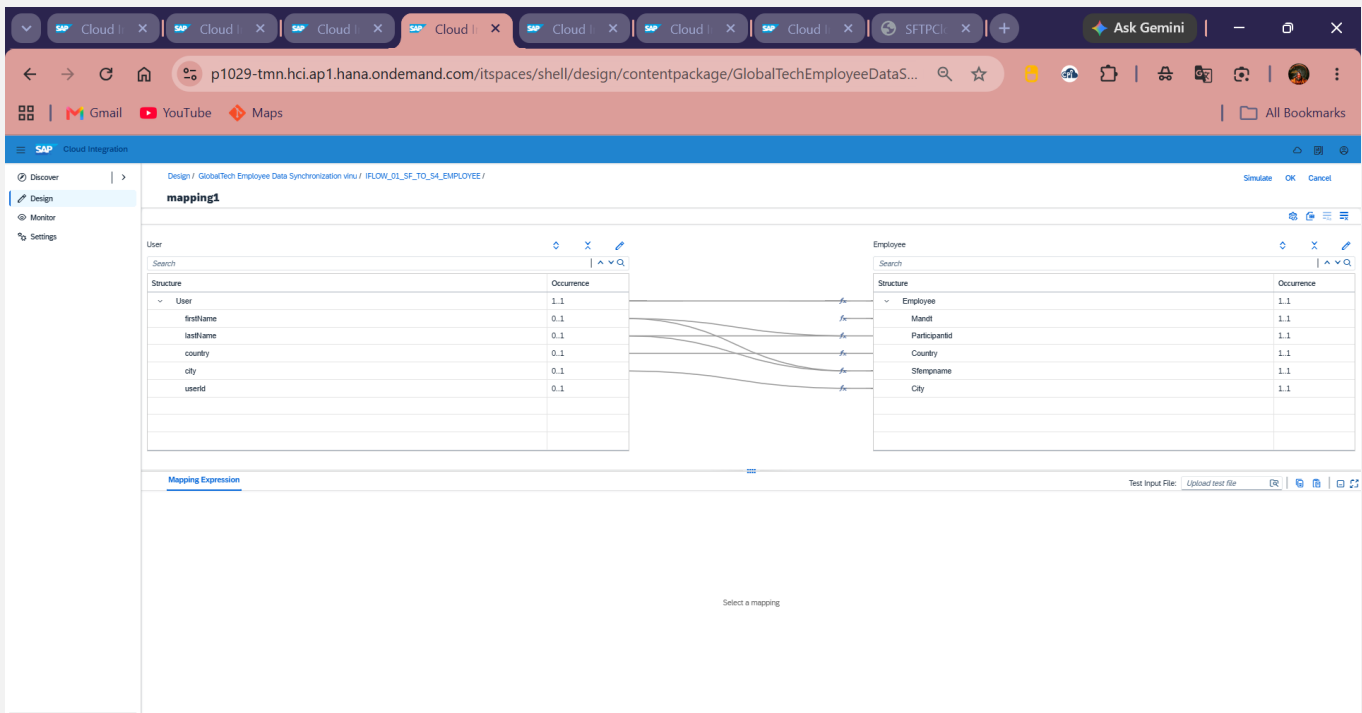
Implementation evidence screenshot 04.

Evidence 05



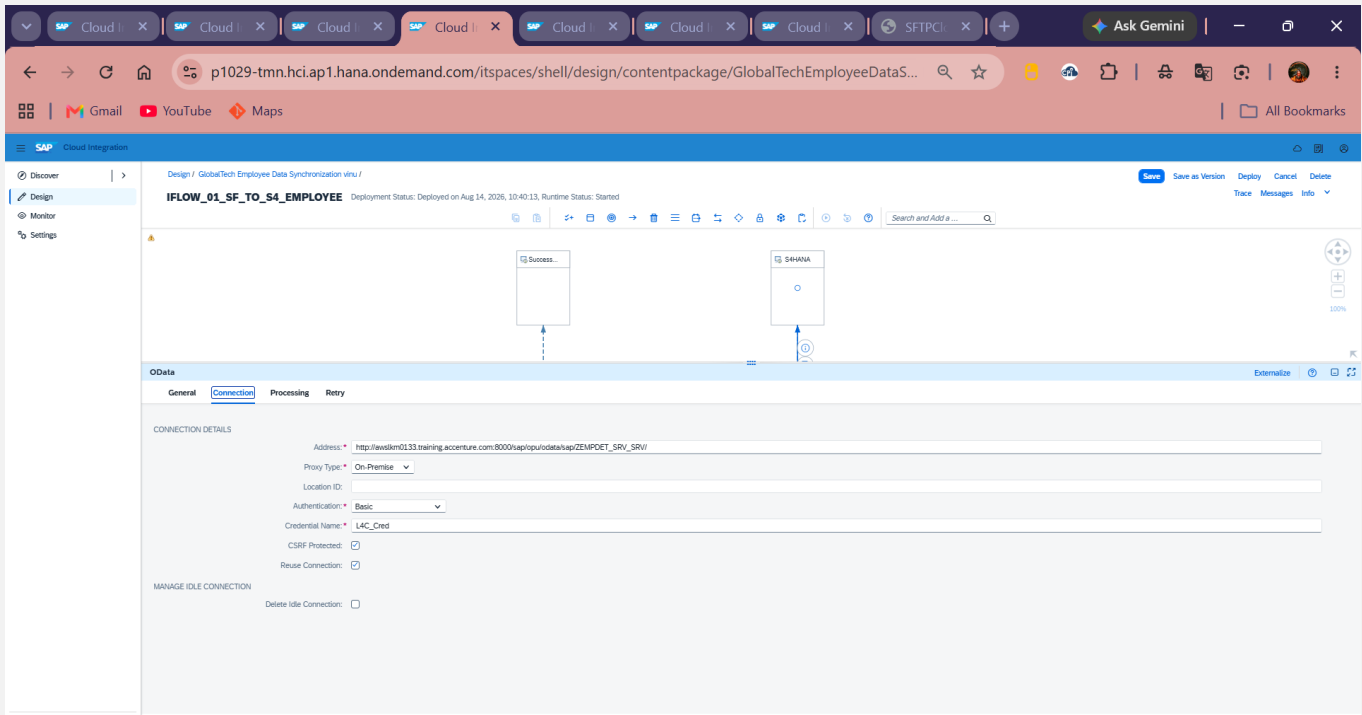
Implementation evidence screenshot 05.

Evidence 06



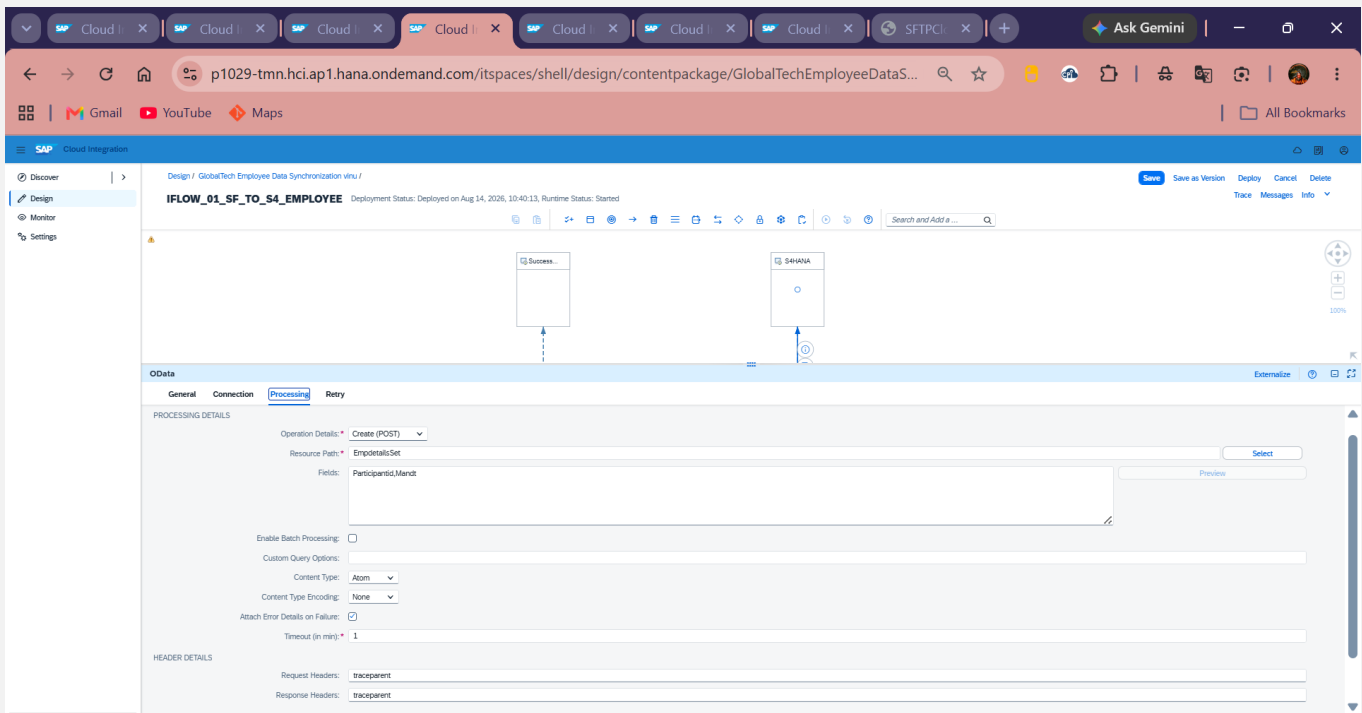
Implementation evidence screenshot 06.

Evidence 07



Implementation evidence screenshot 07.

Evidence 08



Implementation evidence screenshot 08.

Evidence 09

The screenshot displays the SAP Cloud Integration Designer interface. The top navigation bar includes 'Discover', 'Design', 'Monitor', and 'Settings'. The main workspace shows a process diagram for 'IFLOW_01_SF_TO_S4_EMPLOYEE'. Below the diagram, the 'Content Modifier' configuration is visible, with the 'Message Body' tab selected. The configuration includes the following details:

- Type: Expression
- Body: GlobalTech Employee Data Synchronization Failed
- Integration Flow: IFLOW_01_SF_TO_S4_EMPLOYEE
- Error Message: \${exception.message}
- Please check SAP Cloud Integration Message Monitoring.

Implementation evidence screenshot 09.

Evidence 10

The screenshot displays the SAP Cloud Integration Designer interface, showing the 'Mail' configuration for the integration flow 'IFLOW_01_SF_TO_S4_EMPLOYEE'. The 'Mail' configuration is visible, with the 'Connection' tab selected. The configuration includes the following details:

- Address: smtp.gmail.com:587
- Proxy Type: Internet
- Timeout (in ms): 30000
- Protection: STARTTLS Mandatory
- Authentication: Plain User/Password
- Credential Name: vthumail

Implementation evidence screenshot 10.

Evidence 11

SAP Cloud Integration Designer interface showing the design of an integration flow named **IFLOW_01_SF_TO_S4_EMPLOYEE**. The flow is deployed on Aug 14, 2026, 10:40:13, and its runtime status is Started. The design view shows a sequence of steps: Start Timer 1, Content Modifier, and End. An Exception Subprocess 1 is also defined, containing an Error Start 1, Content Modifier, and End 1. The Mail tab is active, displaying the Mail Attributes section with fields for From, To, Cc, Bcc, Subject, and Mail Body. The Mail Body field contains the placeholder `${n.body}`. The Body Mime-Type is set to TextPlain and the Body Encoding is set to UTF-8.

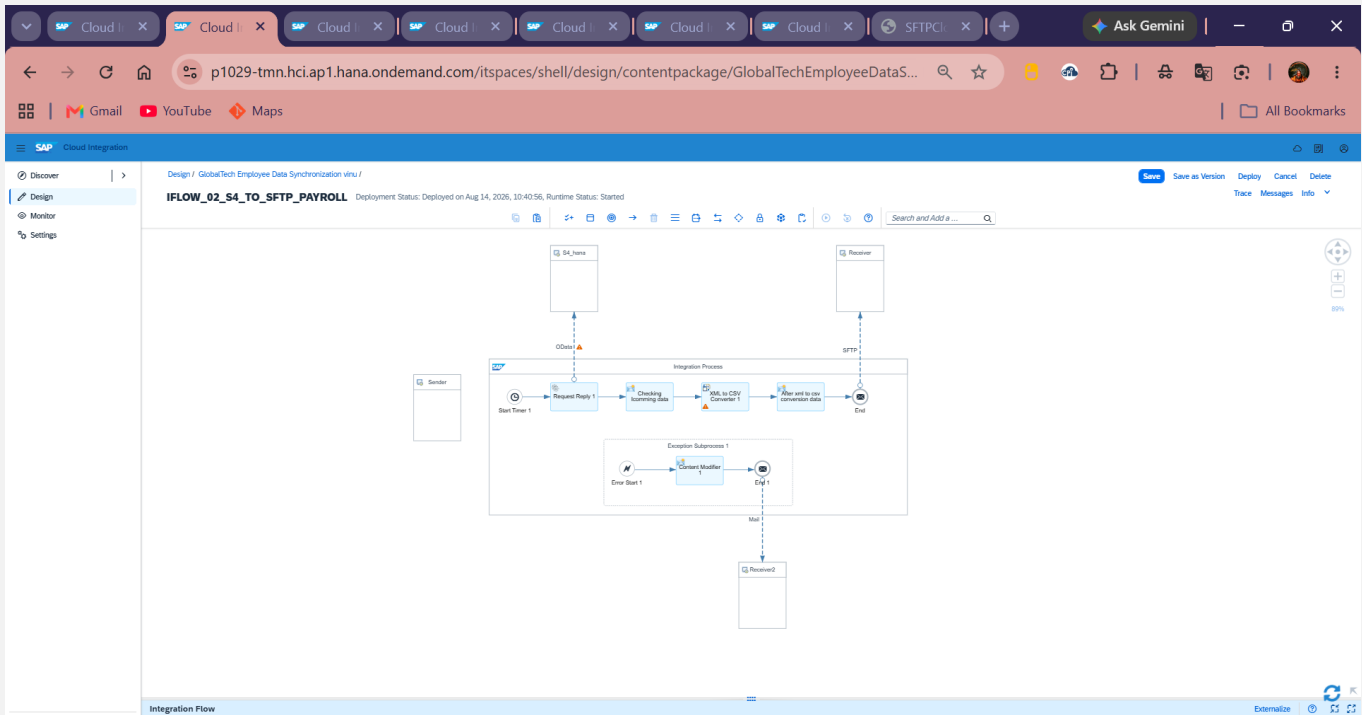
Implementation evidence screenshot 11.

Evidence 12

SAP Cloud Integration Designer interface showing the deployment status of the integration flow **IFLOW_01_SF_TO_S4_EMPLOYEE**. The flow is deployed on Aug 14, 2026, 10:40:13, and its runtime status is Started. The Deployment Status tab is selected, displaying the deployment details. The deployment status is Deployed, and the runtime status is Started. The deployment was performed by P0012764893 on Aug 14, 2026, 10:40:13, and the deployed version is 1.0.0.

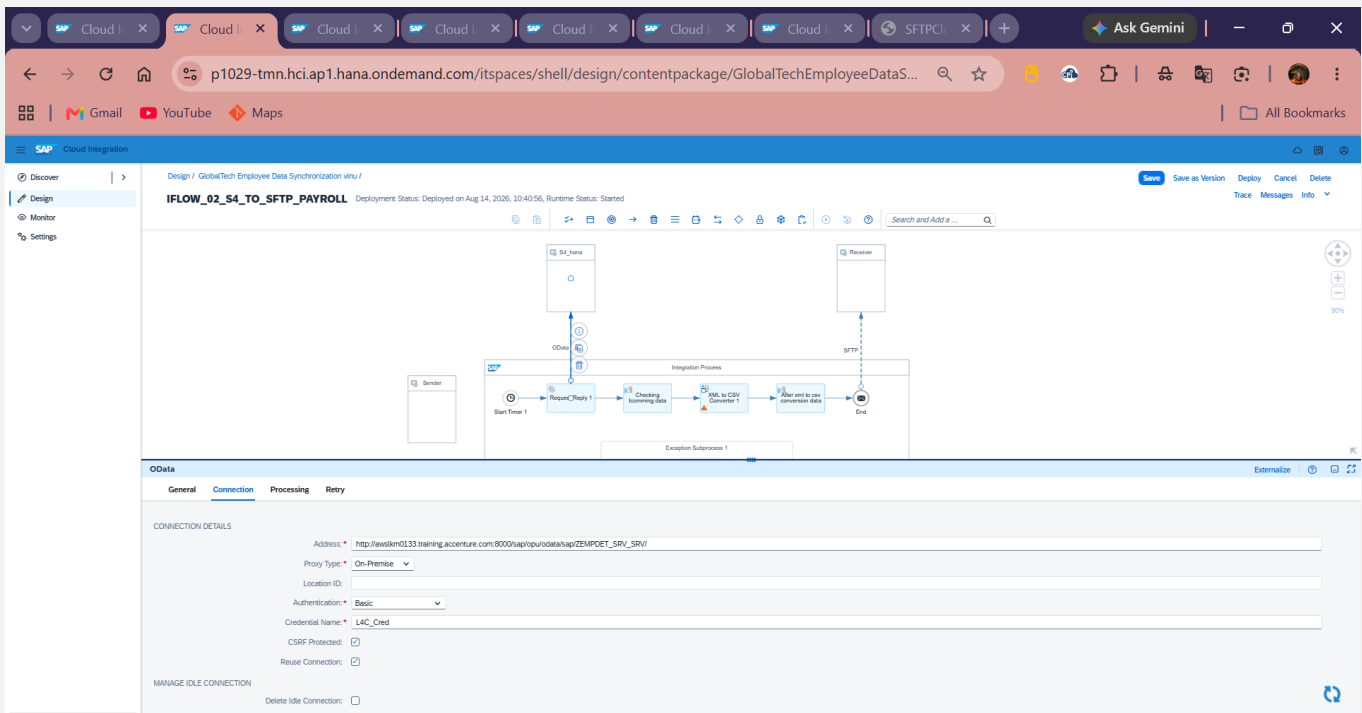
Implementation evidence screenshot 12.

Evidence 13



Implementation evidence screenshot 13.

Evidence 14



Implementation evidence screenshot 14.

Evidence 15

The screenshot displays the SAP Cloud Integration Designer interface for the integration process **IFLOW_02_S4_TO_SFTP_PAYROLL**. The process is designed to synchronize employee data from SAP S/4HANA to an SFTP receiver. The main canvas shows the following components and flow:

- Sender:** SAP S/4HANA.
- Start Timer 1:** Initiates the process.
- Request_Reply 1:** A request-reply connector.
- Checking Incoming data:** A message filter.
- XML to CSV Converter 1:** A message converter.
- After send to sftp conversion data:** A message filter.
- Receiver:** SFTP.

The **Processing** tab is selected, showing the following details:

- Operation Details:** Query (GET)
- Resource Path:** EmpDetailsSet
- Query Options:** \$select=Name,Participant,Stemname,City,Country
- Enable Batch Processing:** ☐
- Custom Query Options:**
- Content Type:** Atom
- Page Size:**

Implementation evidence screenshot 15.

Evidence 16

The screenshot displays the configuration for the **XML to CSV Converter 1** component within the **IFLOW_02_S4_TO_SFTP_PAYROLL** process. The **Processing** tab is selected, showing the following configuration:

- BASIC ELEMENTS:**
 - Path to Source Element in XML:** /EmpDetailsSet/EmpDetails
 - Field Separator in CSV:** Comma(,)
 - CSV Header:** XML_based
- PARENT ELEMENT:**
 - Include Parent Element:** ☐
 - Include Attribute Values:** ☐

Implementation evidence screenshot 16.

Evidence 17

The screenshot displays the SAP Cloud Integration Designer interface for the process **IFLOW_02_S4_TO_SFTP_PAYROLL**. The process flow is as follows:

- Sender** connects to **S4_hana**.
- S4_hana** connects to **Request Reply 1**.
- Request Reply 1** connects to **Checking incoming data**.
- Checking incoming data** connects to **XML to CSV Converter 1**.
- XML to CSV Converter 1** connects to **Filter and to csv conversion data**.
- Filter and to csv conversion data** connects to **End**.

The **Integration Process** is shown with a progress bar at 90%.

The **FILE ACCESS PARAMETERS** section includes:

- Directory: JAEH
- File Name: helloVM.csv
- Append Timestamp: ☐

The **CONNECTION PARAMETERS** section includes:

- Address: us-east-1.sftpcloud.io:22
- Proxy Type: Internet
- Authentication: User Name/Password
- Credential Name: SFTP_vinu
- Timeout (in ms): 10000
- Maximum Reconnect Attempts: 3

Implementation evidence screenshot 17.

Evidence 18

The screenshot displays the SAP Cloud Integration Designer interface for the process **IFLOW_02_S4_TO_SFTP_PAYROLL**. The process flow is as follows:

- Sender** connects to **S4_hana**.
- S4_hana** connects to **Request Reply 1**.
- Request Reply 1** connects to **Checking incoming data**.
- Checking incoming data** connects to **XML to CSV Converter 1**.
- XML to CSV Converter 1** connects to **Filter and to csv conversion data**.
- Filter and to csv conversion data** connects to **End**.

The **Integration Process** is shown with a progress bar at 90%.

The **Content Modifier** section is expanded, showing the **Message Header** tab. The headers are:

Action	Name	Source Type	Source Value	Data Type	Default Value
☐ Action					
☐ Create	Subject	Constant	Subject is empty. Message will be sent without a...		

Implementation evidence screenshot 18.

Evidence 19

The screenshot displays the SAP Cloud Integration Designer interface. The top navigation bar includes 'Discover', 'Design', 'Monitor', and 'Settings'. The main workspace shows an integration process named 'IFLOW_02_S4_TO_SFTP_PAYROLL'. The process flow is as follows: Start Timer 1 → Request Reply 1 → Check and Summing Data → XML to CSV Converter 1 → Filter and to csv conversion date → End. An 'Exception Subprocess 1' is attached to the main process, containing 'Error Start 1' → Content Modifier → End 1. The 'Content Modifier' step is currently selected, showing its configuration in the 'Message Body' tab. The message body is a constant string: '[ERROR] Payroll SFTP Integration Failed. S/HANA to Payroll SFTP Integration failed. Error: \$!exception.message) Stack Trace: \$!exception.stacktrace)'. A text box prompts the user to 'Enter an expression or a constant to form new message body'.

Implementation evidence screenshot 19.

Evidence 20

The screenshot displays the SAP Cloud Integration Designer interface, specifically the 'Mail' configuration section. The top navigation bar is the same as in Evidence 19. The main workspace shows the same integration process. The 'Mail' configuration is open, showing the 'Connection' tab. The 'CONNECTION DETAILS' section contains the following fields: Address: smtp.gmail.com:587, Proxy Type: Internet, Timeout (in ms): 30000, Protection: STARTTLS Optional, Authentication: Plain User/Password, and Credential Name: virtualmail.

Implementation evidence screenshot 20.

Evidence 21

The screenshot displays the SAP Cloud Integration Designer interface. The top navigation bar includes 'Discover', 'Design', 'Monitor', and 'Settings'. The main workspace shows the 'Mail' tab for the integration process 'IFLOW_02_S4_TO_SFTP_PAYROLL'. The process flow is visible, starting with 'Start Timer 1', followed by 'Request Reply 1', 'Checking Running Data', 'XML to CSV Converter 1', and 'Filter and to csv conversion date'. An 'Exception Subprocess 1' is also shown, containing 'Error Start 1' and 'Content Modifier'. The 'Mail' tab is selected, showing the 'MAIL ATTRIBUTES' section with fields for 'From', 'To', 'Cc', 'Bcc', 'Subject', and 'Mail Body'. The 'Subject' field is populated with '\$[header.Subject]' and the 'Mail Body' with '\$[in.body]'. The 'Body Mime-Type' is set to 'TextPlain'.

Implementation evidence screenshot 21.

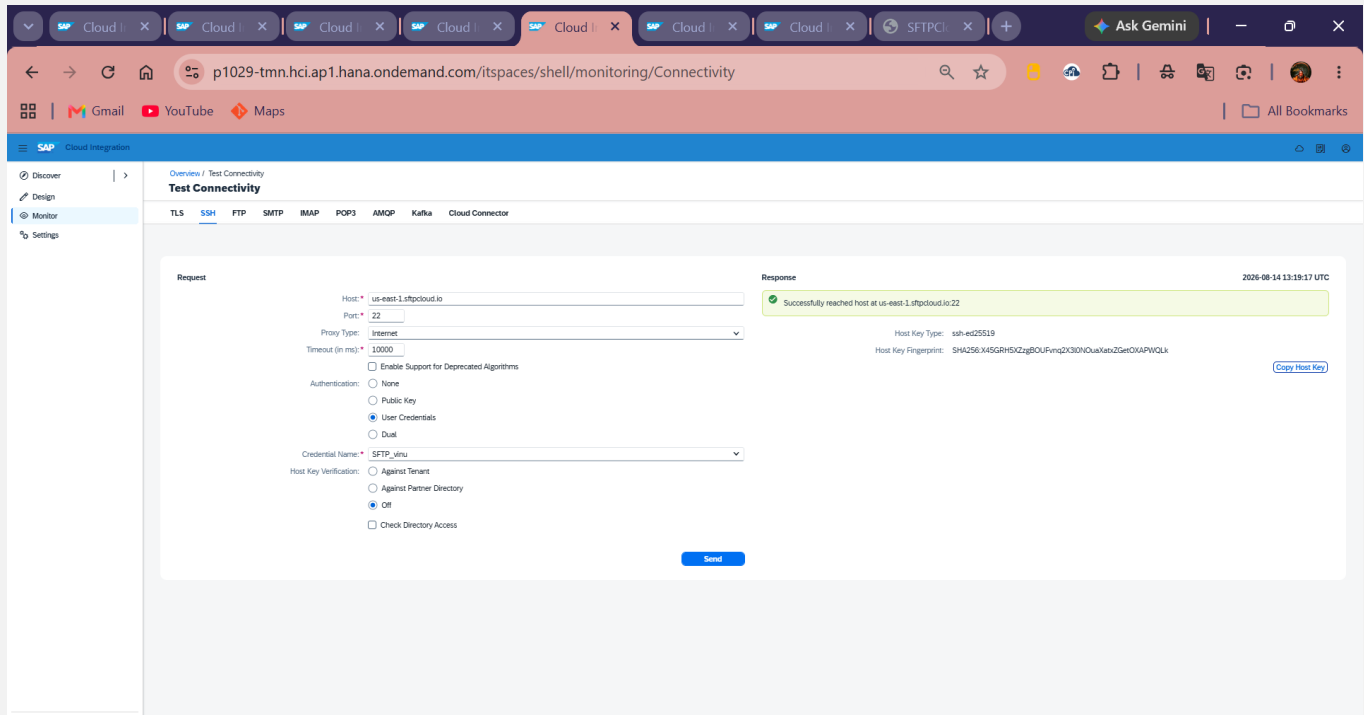
Evidence 22

The screenshot displays the SAP Cloud Integration Designer interface, showing the 'Deployment Status' tab for the integration process 'IFLOW_02_S4_TO_SFTP_PAYROLL'. The process flow is visible, starting with 'Start Timer 1', followed by 'Request Reply 1', 'Checking Running Data', 'XML to CSV Converter 1', and 'Filter and to csv conversion date'. An 'Exception Subprocess 1' is also shown, containing 'Error Start 1' and 'Content Modifier'. The 'Deployment Status' tab is selected, showing the 'Integration Flow' section with a table of deployment details.

Deployment Status	Runtime Status	Deployed By	Deployed On	Deployed Version
Deployed	Started	P2012764983	Aug 14, 2026, 10:40:56	1.0.0

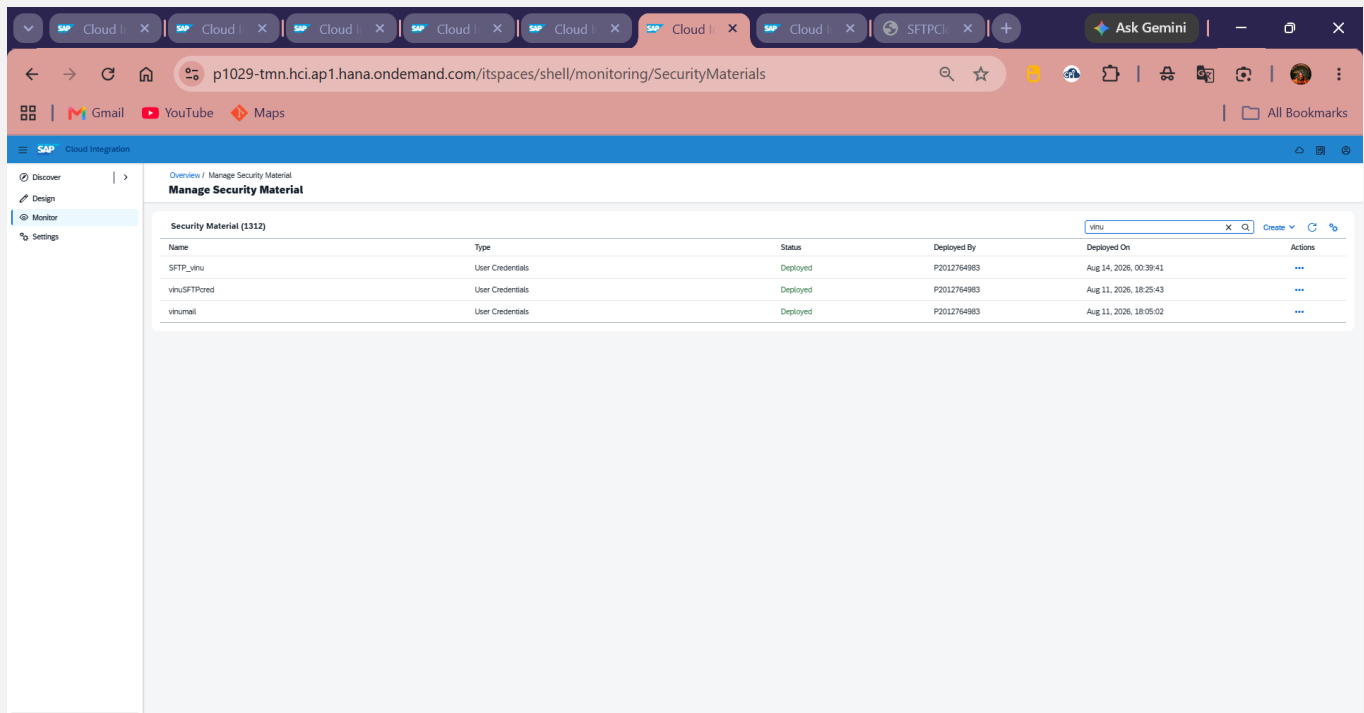
Implementation evidence screenshot 22.

Evidence 23



Implementation evidence screenshot 23.

Evidence 24



Implementation evidence screenshot 24.

Evidence 25

The screenshot shows the SFTPCloud web interface. The browser address bar displays the URL: `app.sftpccloud.io/sftp-instances/5f6cccd-f-ae1-4074-9c85-54a47757d753/users/4f132d6a-6f0a-48b7-9621-890a...`. The interface has a top navigation bar with tabs for 'Overview', 'Users', 'File manager', 'Event listeners', and 'Logs'. The 'Users' tab is active. On the left, there is a sidebar with a user profile for 'Store' (Root user) and a list of settings: 'Connection details', 'Authentication', 'Public Keys', 'Permissions', 'Allowed/denied IPs', 'Quotas', and 'Logs'. The 'Connection details' section is expanded, showing the following information:

Host	Username
us-east-1.sftpccloud.io	Store

Password	SFTP Port
*****	22

FTP Port	Server Fingerprint
21	Click for details

Below the connection details, there is a 'Guides' section with links to 'How to connect via SFTP' and 'How to connect via FTPS'. A 'Delete user' button is visible in the top right corner of the user details section.

Implementation evidence screenshot 25.

Evidence 26

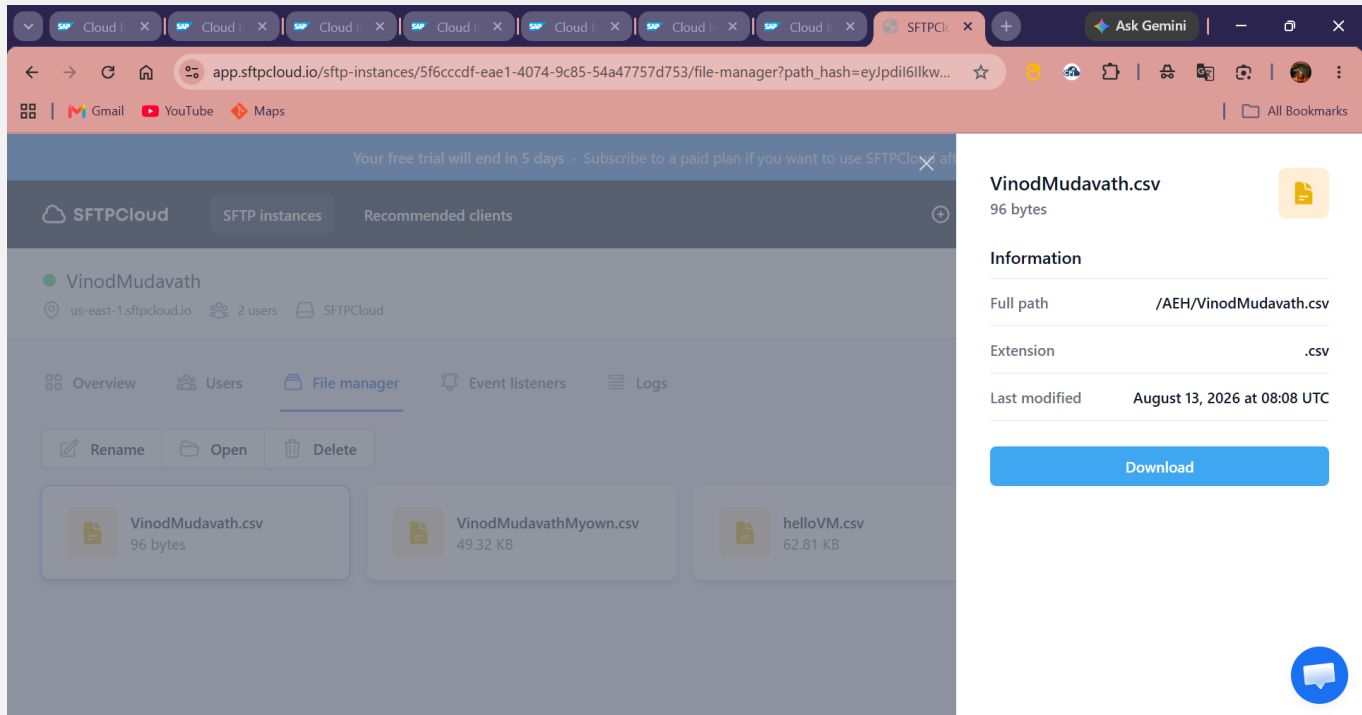
The screenshot shows the SFTPCloud web interface. The browser address bar displays the URL: `app.sftpccloud.io/sftp-instances/5f6cccd-f-ae1-4074-9c85-54a47757d753/file-manager?path_hash=eyJpdil6lkw...`. A blue banner at the top states: 'Your free trial will end in 5 days · Subscribe to a paid plan if you want to use SFTPcloud after that →'. The interface has a top navigation bar with tabs for 'SFTPCloud', 'SFTP instances', and 'Recommended clients'. The 'SFTP instances' tab is active. Below the navigation bar, there is a header for the user 'VinodMudavath' (us-east-1.sftpccloud.io) with '2 users' and 'SFTPcloud' listed. The 'File manager' tab is active. The directory listing shows the following files:

File Name	Size
VinodMudavath.csv	96 bytes
VinodMudavathMyown.csv	49.32 KB
helloVM.csv	62.81 KB

Buttons for 'Directory logs' and 'New directory' are visible in the top right corner of the file manager section.

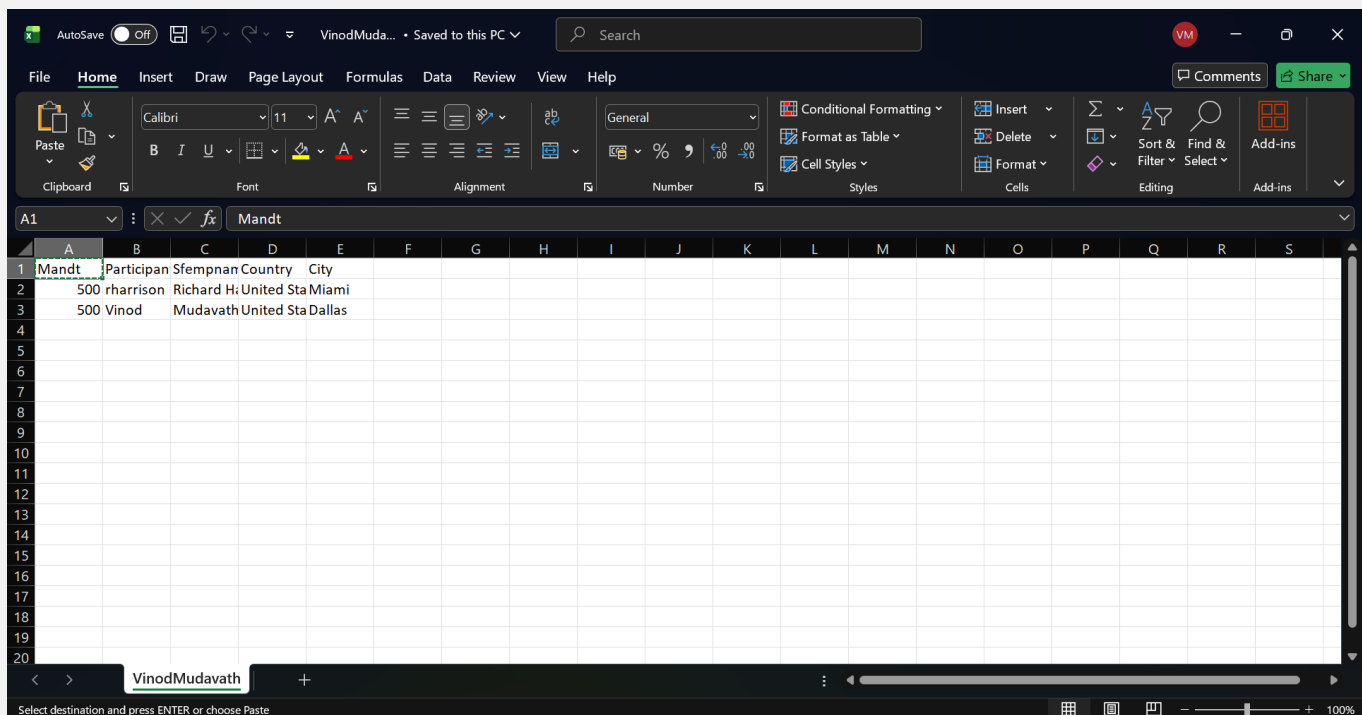
Implementation evidence screenshot 26.

Evidence 27



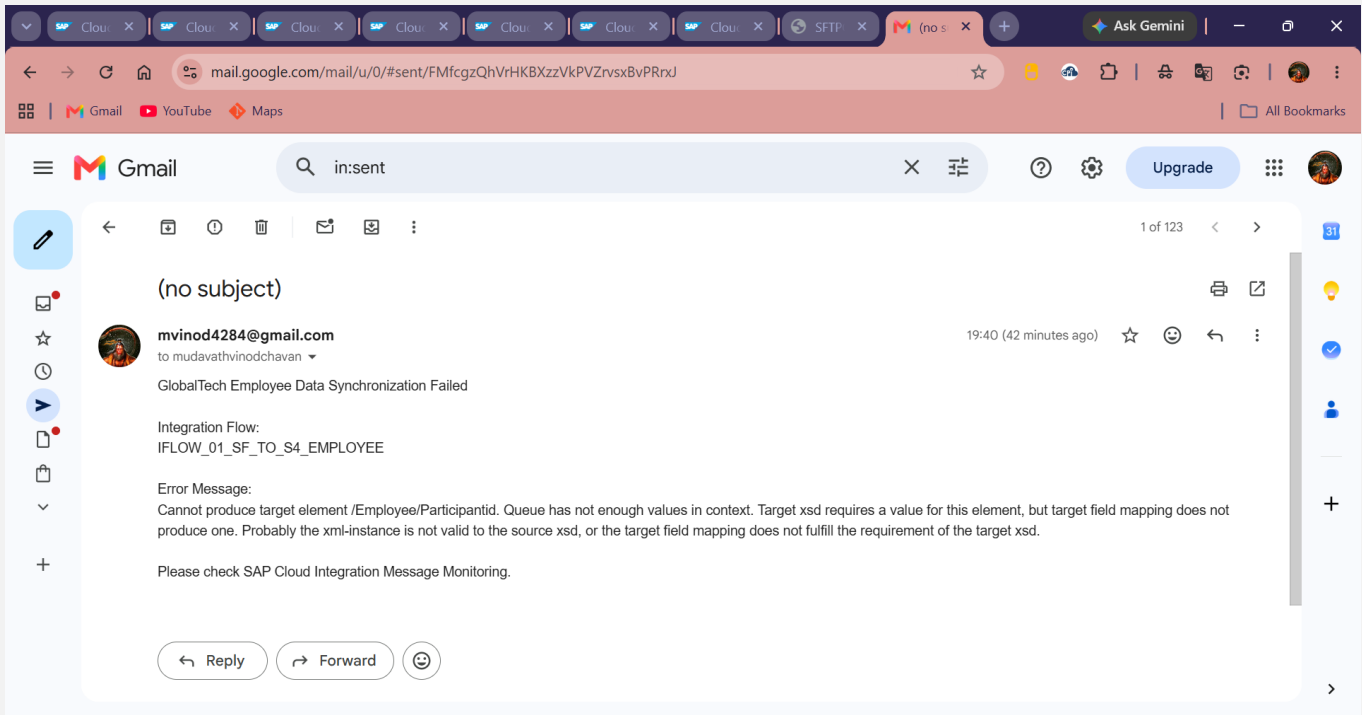
Implementation evidence screenshot 27.

Evidence 28



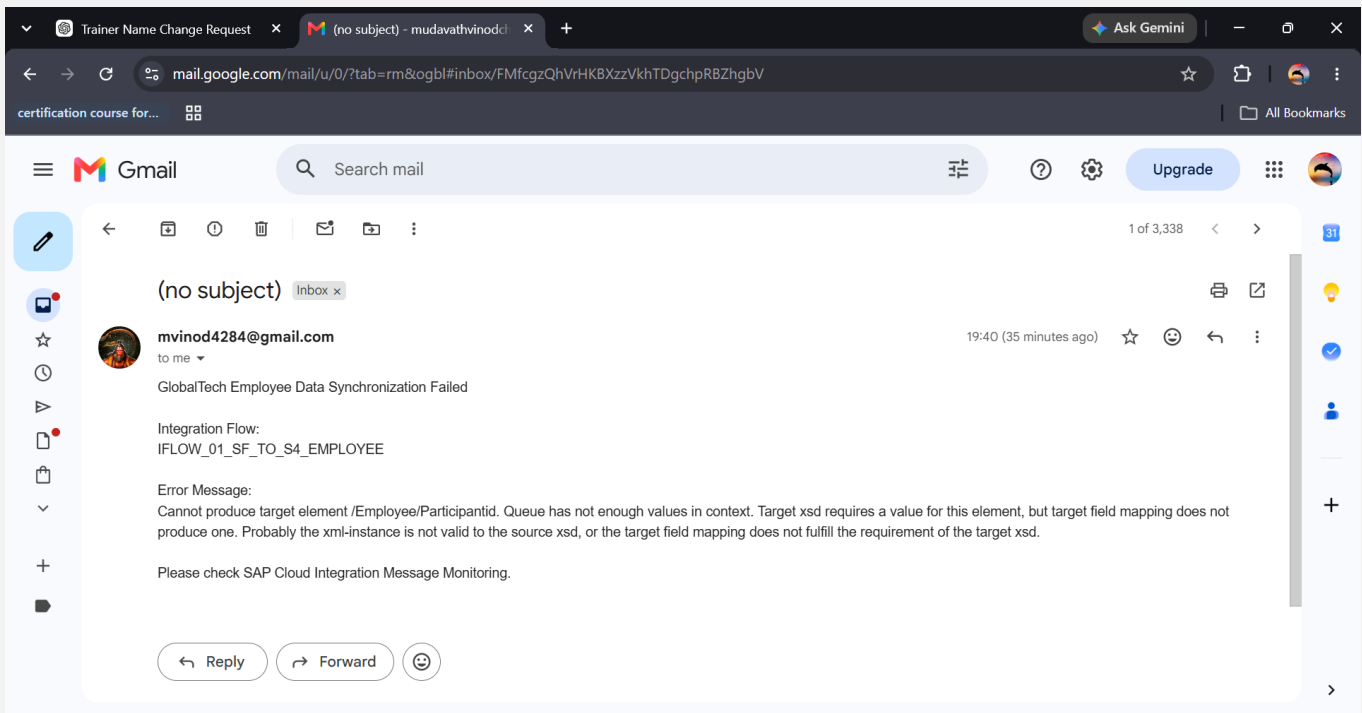
Implementation evidence screenshot 28.

Evidence 29



Implementation evidence screenshot 29.

Evidence 30

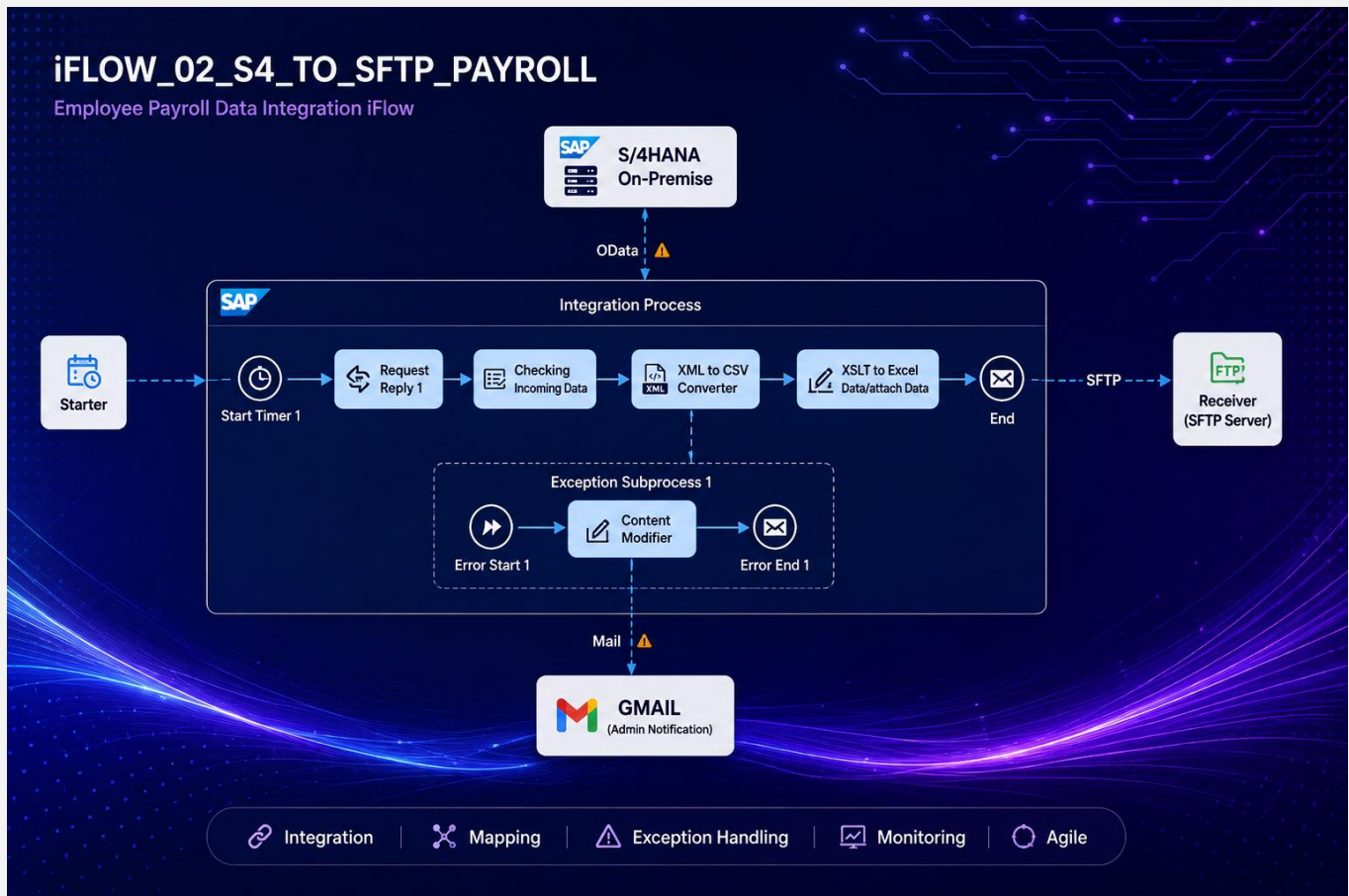


Implementation evidence screenshot 30.

Evidence 31



Implementation evidence screenshot 31.



Implementation evidence screenshot 32.

